

IEEE Sri Lanka Challenge Sphere 2025

Problem Statement - Sri Lanka Arduino Challenge 2025

1. About the Competition

The Sri Lanka Arduino Challenge 2025 fosters innovation, collaboration, and hands-on learning in electronics, telecommunications, and embedded systems. It is organized by the IEEE Industrial Electronics Society (IES) Student Branch Chapter of SLTC Research University, collaborating with IEEE Sri Lanka Section, Challenge Sphere 2025.

1.1. Overview

The Arduino Challenge invites students to develop innovative solutions using Arduino technology, enhancing their practical skills and preparing them for the international stage.

1.2. Purpose/Objectives

- Encourage participants to address real-world issues through innovative solutions using Arduino technology.
- Develop critical thinking, systems design, and troubleshooting skills essential for embedded systems and electronics innovation.
- Unite students from diverse academic disciplines to work as cohesive teams, blending expertise in electronics, software, and systems engineering to create innovative Arduino-based solutions.
- Prepare students to confidently participate in the IEEE SSCS Arduino Contest (Global Contest) 2025 by building their technical skills, project experience, and innovation mindset for success on the international stage.

2. Rules and Regulations

2.1. Eligibility Criteria

Teams of up to three students from across Sri Lanka are eligible to participate in the competition. The following categories of students are allowed:

- School Students: Students enrolled in any Sri Lankan school, with a school completion date on or after 31st December 2025.
- University Undergraduates: Students currently enrolled in any Sri Lankan university, with a graduation date on or after 31st December 2025.

2.2. Guidelines

Project Theme: Assisting People with a Disability. Each team must select a specific focus under the main theme of "Assisting People with a Disability". Begin by conducting thorough research to identify a real-world problem that affects individuals with disabilities and design an Arduino-based solution to address it.

Suggested Sub-Themes:

- Bionic limbs for physical impairments
- Visual alert systems for hearing impairments
- Obstacle detection and navigation tools for vision impairments
- Accessible smart devices for mobility challenges
- Other relevant and impactful sub-themes are welcome

Entry Submission Guidelines:

Once your project is complete, submit a 3-minute demo video along with a brief description. Videos may be submitted in any language. This submission is mandatory and serves as part of your team's official registration for the Arduino Challenge.

- Submission will be via Google Form, which can be edited until the deadline
- Only one (1) entry per team is allowed

- Submission Deadline: 23rd July 2025

Your Demo Video Must Include:

- A team introduction
- The name of the school or university and the country you are representing
- A demonstration of your project in action
- One challenge your team faced and how you overcame it

Material:

The participants are **only allowed** to use the following electronic components in their projects for the competition:

1. (1) Arduino-compatible microcontroller
2. (1) Breadboard/ Perfboard/ PCB
3. (1) Power Cable or Battery
4. (Infinity amount) Wires
5. (Infinity amount) Resistors & capacitors
6. (Infinity amount) Inductors
7. (4) Diodes
8. (4) Transistors
9. (4) Potentiometers
10. (1) LED SCREEN
11. (1) Number Pad
12. (4) Push Buttons
13. (2) Switch
14. (4) Relay Switches
15. (4) DC Motors and Drivers
16. (4) Servo Motors and Drivers
17. (1) Buzzer Motor and Driver
18. (1) Microphone and Recorder
19. (1) Speaker and Audio Driver
20. (1) WIFI or Bluetooth Module
21. (16) LEDs
22. (1) Distance Sensor
23. (1) Light Sensor
24. (1) IR Transmitter and Receiver
25. (1) Temperature Sensor
26. (1) Humidity and Moisture Sensor
27. (1) Motion Sensor
28. (1) Gyroscope and Accelerometer
29. (1) Force or Pressure Sensor
30. (1) Flex (bending) Sensor
31. (1) RFID Tag and Receiver
32. (1) Hall Effect (Magnet) Sensor
33. (1) GPS Module
34. (1) No built-in Intelligence

- But snaz up your project with whatever non-electronic materials you wish!
- Build a nice enclosure (if applicable) to house your project.

Deliverables:

The final project must have:

- Arduino-based final project prototype
- A demonstration with (at least) 5 slides (maximum 10 slides)
- 3-minute demo video showcasing the functionality and purpose of the project

Demo & Exhibition:

All teams are required to present a 10-minute live demonstration of their innovation during the final competition and exhibition on 7th August 2025. The final competition will be an in-person event at SLTC, Trace Expert City, Maradana, Colombo 10.

Awards & Opportunities :

- The top 3 teams will be awarded cash prizes and winner certificates, while each member of the remaining 7 finalist teams will receive certificates of achievement in recognition of their outstanding efforts and participation.
- The Top 10 teams will be invited to attend follow-up sessions aimed at helping them prepare and submit their projects for the IEEE SSCS Arduino Contest (Global Contest) 2025

2.3. Judging Criteria

Your deliverables and the final showcase will be evaluated based on the following criteria;

- Idea of the project: 10%
- Impact of the project on the Climate: 15%
- Technical feasibility: 10%
- Marketing feasibility: 5%
- Scalability of the project: 10%
- Final demonstration + Slides of the project: 50%

After the submission deadline, the **top 10 teams** will be selected and invited to participate in the **in-person competition and exhibition**.

A panel of judges will evaluate the live demonstrations and **select up to three teams** with the most outstanding projects to be recognized as **winners**.

2.4. Timeline

Important dates:

- Awareness Session and Theme Reveal

Starting Date: 22/06/2025

Date of Completion: 22/06/2025

- Registrations Open:

Starting Date: 23/06/2025

Date of Completion: 23/06/2025

- Registrations Close:

Starting Date: 23/07/2025

Date of Completion: 23/07/2025

- Online Workshop 1

Starting Date: 28/06/2025

Date of Completion: 28/06/2025

- Online Workshop 2

Starting Date: 03/07/2025

Date of Completion: 03/07/2025

- Submission Deadline

Starting Date: 23/07/2025

Date of Completion: 23/07/2025

- Finalist Announcement

Starting Date: 26/07/2025

Date of Completion: 26/07/2025

- Exhibition and Competition

Starting Date: 07/08/2025

Date of Completion: 07/08/2025

Program Plan (Descriptively):

- **Opening Ceremony** – The opening ceremony, organized by the IEEE Sri Lanka Educational Activities Sub Committee under the leadership of Eng. Dilini Ekanayaka was held on April 7th, 2025.
- **Awareness Webinar on Arduino and Theme Reveal** – The inaugural event will provide participants with an overview of the Sri Lanka Arduino Challenge Competition, introduce them to Arduino technology and its wide range of applications, and conclude with the official reveal of the competition theme, under which participants will develop their Arduino-based innovative projects.
- **Arduino Online Workshop Series** – This online hands-on training series, comprising two workshops, is designed to equip participants with practical skills and foundational knowledge in Arduino programming and electronics.
- **Physical Exhibition and Competition** – The final phase of the program will feature a physical exhibition and competition, providing participants with a platform to present their Arduino-based innovations. Projects will be evaluated by a panel of judges, and the most outstanding projects will be recognized and awarded as winners.
- **Closing Ceremony** – The closing ceremony, also organized by the IEEE Sri Lanka Educational Activities Sub Committee under the leadership of Eng. Dilini Ekanayaka will be held in the first week of August.

Note:

The opening and closing ceremonies are general events conducted for participants from various competitions organized by the IEEE Sri Lanka Section, Educational Activities Committee; therefore, there is no major contribution from our organizing committee

2.5. Agreement to Official Rules

By participating in this contest, all entrants agree to comply with the rules and guidelines established by the SLTC Organizing Committee. The SLTC OC reserves the right to modify these official rules at any time and for any reason. All decisions made by the SLTC OC regarding the contest will be final and binding at the sole discretion of the committee.

3. Logistics

3.1. Registration Process

Students can register for the competition by completing the registration form from the 23rd of June onwards.

3.2. Location

The final competition will be an in-person event at SLTC, Trace Expert City, Maradana, Colombo 10.

3.3. Awards

Top 3 Winners:

- Cash Prizes: Awarded to the top 3 winning teams.
- Winner's Certificate: Each member of the top 3 teams will receive a certificate recognizing their achievement.
- Each member of the remaining 7 teams will also receive valuable certificates.

Guidance for IEEE SSCS Arduino Contest:

- The Top 10 teams will be invited to attend follow-up sessions aimed at helping them prepare and submit their projects for the [IEEE SSCS Arduino Contest](#) (Global Contest) 2025.